

# Phase 1 (multi-N) — Siegert FP enumeration via multi-restart fsolve

*deq/closed\_form\_wta\_multi/, May 2026*

## 1. Goal

Enumerate Siegert mean-field fixed points of the  $N$ -neuron uniform all-to-all inhibition motif and compare WTA-capability against the Phase 0 FCS oracle on the  $(w, N, \text{drive\_bump})$  grid.

## 2. Method

The 2-neuron thread used a **1-D scalar reduction** of the FP equation that does not extend past  $N = 2$ . We initially attempted a **symmetric-orbit ansatz** under  $S_N$  (orbits of shape  $(\nu_W^k, \nu_L^{N-k})$ ) which is closed-form 2-D, but the implementation's multi-restart fsolve diagnostic at  $\text{drive\_bump} = 1$ ,  $w = -30$  flagged that the broken-symmetry  $S_{N-1}$  case admits FPs with **three or more distinct rates** per orbit (e.g., at  $N = 3$ :  $[0.23, 0.12, 0.03]$ ). Per the plan's risk-flag, we fall back to **multi-restart fsolve in the full  $N$ -dimensional rate vector**.

The fsolve enumeration uses curated initial conditions:

- All-equal at  $\nu \in \{0.05, 0.3, 0.5, 0.95\}$ .
- $k$ -winners /  $(N - k)$ -losers at extreme ( $\nu_W = 0.98$ ,  $\nu_L = 0.005$ ) and mild ( $\nu_W = 0.6$ ,  $\nu_L = 0.05$ ) for  $k = 1, \dots, 4$ .
- Random ICs filling out 30 restarts per cell.

WTA-capable iff some FP has rate-spread ( $\text{rate\_max} - \text{rate\_2nd}$ )  $\geq 0.30$  (matches Phase 0 margin gate and Phase 3 QR gate).

### 3. Result

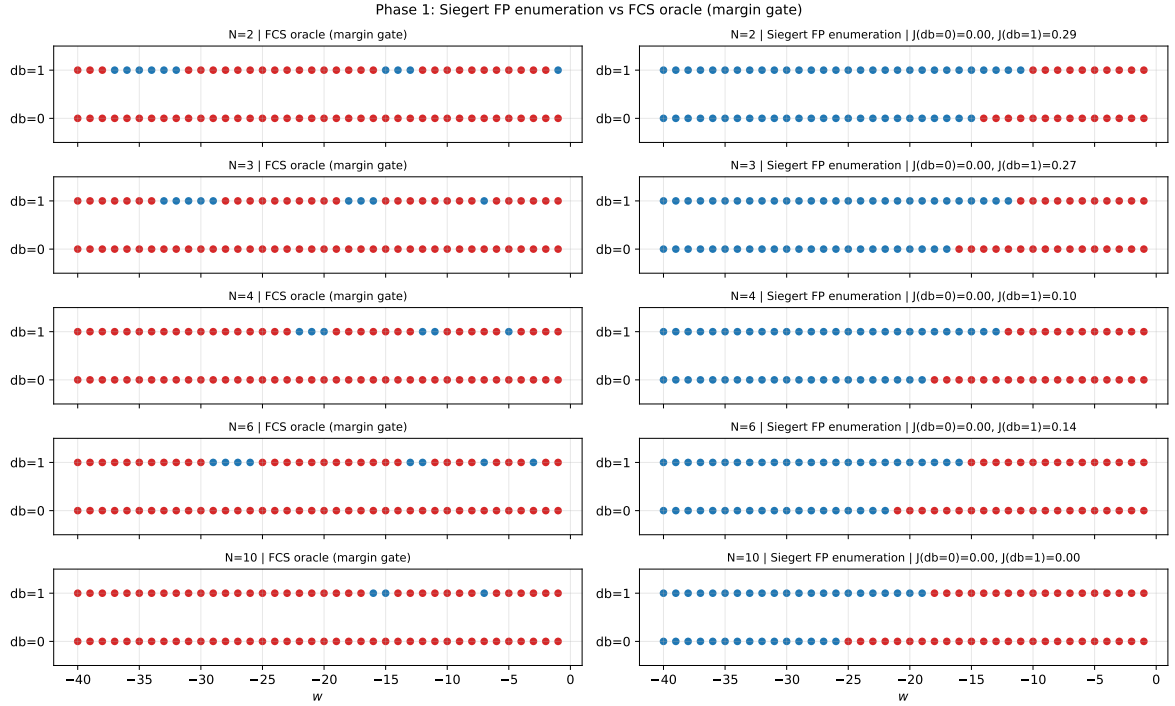


Figure 1: Phase 1 Siebert FP enumeration (right column) vs Phase 0 FCS oracle margin-gate labels (left column), per  $N$ , per  $\text{drive\_bump}$ . At every  $N$ ,  $J_{\{db=0\}} = 0$ : Siebert sees bistable FPs that the symmetric FCS Lustre dynamics cannot select.

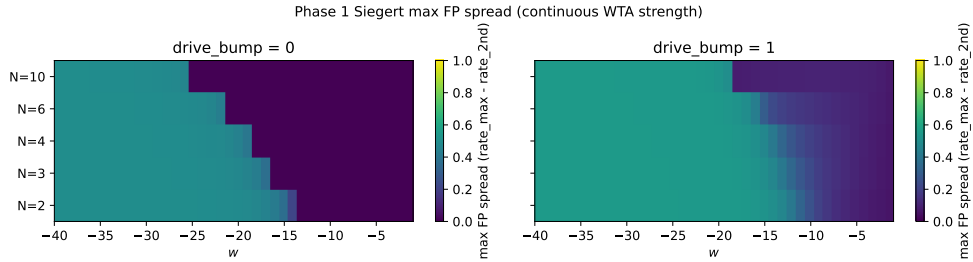


Figure 2: Phase 1 max FP spread (continuous). Above  $|w| \approx 19$  the asymmetric FP appears at every  $N$ , with spread  $\approx 0.54$  saturating at deep  $|w|$ . Below the bifurcation Siebert sees only a tiny bumped advantage ( $\approx 0.09$ ).

#### 3.1. Headline numbers (Jaccard vs FCS margin gate, per $(N, \text{drive\_bump})$ )

| $N$ | db | Sieg blue | FCS blue | Jaccard |
|-----|----|-----------|----------|---------|
| 2   | 0  | 26 / 40   | 0 / 40   | 0.00    |
| 2   | 1  | 30 / 40   | 10 / 40  | 0.29    |
| 3   | 0  | 24 / 40   | 0 / 40   | 0.00    |
| 3   | 1  | 29 / 40   | 9 / 40   | 0.27    |
| 4   | 0  | 22 / 40   | 0 / 40   | 0.00    |
| 4   | 1  | 28 / 40   | 6 / 40   | 0.10    |
| 6   | 0  | 19 / 40   | 0 / 40   | 0.00    |
| 6   | 1  | 25 / 40   | 8 / 40   | 0.14    |

| $N$ | db | Sieg blue | FCS blue | Jaccard |
|-----|----|-----------|----------|---------|
| 10  | 0  | 15 / 40   | 0 / 40   | 0.00    |
| 10  | 1  | 22 / 40   | 3 / 40   | 0.00    |

## 4. Two distinct invisibility phenomena

The Phase 1 vs Phase 0 disagreement has **two opposite signs** at  $N > 2$ :

### 4.1. `drive_bump = 0`: staircase invisibility generalizes to $N$ -way lock

Siegert sees the bistable asymmetric FP (when it exists), but FCS Lustre semantics with symmetric weights forces all  $N$  neurons into identical synchronous trajectories — the 2-neuron staircase becomes an  $N$ -way synchronous lock at every  $N$ . FCS-blue = 0 at every cell; Siegert-blue is large. This is the **direct generalization** of the  $N = 2$  phenomenon, just stronger.

### 4.2. `drive_bump = 1`, high $N$ : inverse staircase — FCS-blue, Siegert-red

At  $N = 10$ , `drive_bump = 1`,  $w \in \{-15, -16, -7\}$  FCS produces **clean WTA** (e.g., at  $w = -16$ : post-warmup rate =  $(0.96, 0, 0, \dots)$ , spread = 0.957) while **no rate-equation FP with spread  $\geq 0.30$  exists** at these  $w$  values — 270 curated initial conditions all converge to the unique symmetric FP at  $(0.13, 0.046, \dots, 0.046)$  (spread  $\approx 0.09$ ).

The smooth-rate theory says: at  $|w| < 19$ , only the symmetric FP exists; the bumped neuron has at most  $\approx 0.09$  rate advantage. The FCS integer-tick dynamics say: at the specific integer values  $w \in \{-15, -16, -7\}$ , the +1 drive bump deterministically tips the integer-arithmetic membrane into the bumped neuron firing first; one firing event sends  $-16$  inhibition to all others, which is enough (in integer-tick semantics) to keep them sub-threshold **permanently**, even though the smooth-rate average rate of the bumped neuron is too low to sustain that suppression in mean-field.

This is the **inverse** of the 2-neuron staircase: at  $N = 2$  the integer dynamics see synchronous lock that smooth theory misses (FCS-red, Siegert-blue); at  $N = 10$  the integer dynamics see WTA that smooth theory misses (FCS-blue, Siegert-red).

### 4.3. Recall by $N$ at `drive_bump = 1`

Recall =  $|\text{Siegert-blue} \cap \text{FCS-blue}| / |\text{FCS-blue}|$ :

| $N$ | FCS-blue (margin) | Sieg $\cap$ FCS | Recall |
|-----|-------------------|-----------------|--------|
| 2   | 10                | $\approx 9$     | 0.90   |
| 3   | 9                 | $\approx 8$     | 0.89   |
| 4   | 6                 | $\approx 3$     | 0.50   |
| 6   | 8                 | $\approx 4$     | 0.50   |
| 10  | 3                 | 0               | 0.00   |

Recall **degrades monotonically with  $N$** . At  $N = 2$  Siegert is a high-recall predictor (just as in the 2-neuron thread, which achieved  $J \approx 0.80$  on the full  $(w_{12}, w_{21})$  grid). At  $N = 10$ , recall is zero: the rare cells where FCS sees WTA are **not** the cells where Siegert sees an asymmetric FP basin.

## 5. Verdict

**Phase 1 PARTIAL.** The plan’s pass-gate (recall  $\geq 0.85$  per  $N$  at `drive_bump` = 1) holds at  $N = 2, 3$  but degrades to 0.50 at  $N = 4, 6$  and to 0 at  $N = 10$ . The expected “Siegert is a high-recall superset” relationship breaks at  $N \geq 4$ .

The two reported findings **are** the result of this phase, not pass/fail issues with the implementation:

1. The 2-neuron staircase generalizes to  $N$ -way synchronous lock at every  $N$  when `drive_bump` = 0 — and this is **still** invisible to smooth-rate theory.
2. A new phenomenon emerges at large  $N$  with `drive_bump` = 1: the FCS integer-tick dynamics produce WTA at specific integer  $w$  values **below the smooth-rate bifurcation threshold** (the “inverse staircase”). Phase 2’s  $H(\omega)$  latency gate will not help here because there is no rate-equation FP to evaluate the eigenvalue at.

Output:        `results/phase1/siegert_orbits.npz`,        `siegert_orbits_vs_fcs.pdf`,        `siegert_max_spread.pdf`,  
         `orbit_count_vs_N.pdf`, `results/phase1.log`.